TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

BE Programme | Year III / Part II

EMBEDDED SYSTEM (CT 655)

QUESTION BANK WITH ANSWERS

Chapter 8: IC Technology

COURSE SYLLABUS AT A GLANCE

Unit	Topic	Hours	Marks
1	Introduction to Embedded System	3	4
2	Hardware Design Issues	4	8
3	Software Design Issues	6	8
4	Memory	5	8
5	Interfacing	6	8
6	Real-Time Operating System (RTOS)	8	12
7	Control System	3	8
8	IC Technology	3	8
9	Microcontrollers in Embedded Systems	3	8
10	VHDL	4	8
	Total	**45**	**80**

Chapter 8 Topics (3 Hours | 8 Marks):

- 8.1 Full-Custom (VLSI) IC Technology — Design Steps, Advantages, Disadvantages
- 8.2 IC Manufacturing Process — Silicon Wafer to Packaged Chip
- 8.3 Photolithography — Process Steps, Types (Contact, Projection, EUV, E-beam)
- 8.4 Semi-Custom (ASIC) IC Technology — Gate Array, Standard Cell
- 8.5 Programmable Logic Device (PLD) IC Technology — SPLD, CPLD, FPGA
- 8.6 Comparison: Full-Custom vs Semi-Custom vs PLD
- 8.7 CMOS Technology — NAND, NOR Gate Design using CMOS

Chapter 8: IC Technology

Q1. What are the steps involved in manufacturing of an IC? Show the various steps of the photolithography process using appropriate diagrams.	**2082 Kartik (Back) | 2081 Ashwin (Back) | 2080 Ashwin (Back) | 2075 Bhadra (Back) [3+5]**

ANSWER

IC Manufacturing — Overview:

```
Silicon Sand → [Purification] → [Czochralski Growth] → Silicon Ingot
     → [Wafer Slicing] → [Polishing] → Silicon Wafer
     → [Epitaxial Growth] → [Repeated: Photolithography + Etch + Dope +
Deposit]
     → [Metallization] → [CMP] → [Wafer Test] → [Dicing] → [Packaging] →
IC
```

Detailed IC Manufacturing Steps:

1. Silicon Ingot Growth (Czochralski Process): Ultra-pure polycrystalline silicon is melted in a quartz crucible. A seed crystal is dipped in and slowly pulled while rotating, growing a single-crystal cylindrical silicon ingot (diameter 200mm or 300mm).
2. Wafer Slicing and Polishing: The ingot is sliced into thin wafers (~0.5–1 mm thick) using a diamond wire saw. Wafers are lapped and chemically mechanically polished (CMP) to a mirror-smooth surface (roughness < 0.1 nm).
3. Epitaxial Layer Growth: A thin (1–10 μm), precisely doped silicon layer is grown on the wafer surface by Chemical Vapor Deposition (CVD). This provides a clean, controlled starting layer for transistor fabrication.
4. Photolithography: Pattern transfer from photomask to wafer surface using UV/EUV light and photoresist. Repeated for every layer (~20–30 cycles in modern chips). (See detailed steps below.)
5. Etching: Remove exposed material to create circuit features. Wet etching uses chemical solutions (HF for SiO2). Dry/Plasma etching uses ionized gas (more precise, anisotropic — used for sub-micron features).
6. Doping (Ion Implantation): Dopant ions (Boron for p-type, Phosphorus/Arsenic for n-type) are accelerated and implanted into exposed silicon regions to form source, drain, and well regions of MOSFETs.
7. Thin Film Deposition: Insulating layers (SiO2, Si3N4) by CVD; conducting layers (polysilicon for gates, tungsten for vias, copper for interconnects) by CVD or PVD (sputtering).
8. Chemical Mechanical Planarization (CMP): Polishes each deposited layer flat to allow accurate multi-level metallization (modern chips have 10–15 metal layers).
9. Metallization: Deposit and pattern metal interconnect layers (aluminum in older processes, copper in modern) to connect all transistors and form circuit wiring.

10. Wafer Testing (Probe Test): Automated probe stations electrically test every die on the wafer. Failing dies are inked/marked.
11. Dicing: Diamond saw or laser dices the wafer into individual chips (dies).
12. Packaging: Die is bonded to a lead frame or substrate; bond wires connect die pads to package pins; encapsulated in plastic (DIP, QFP, BGA) or ceramic.
13. Final Testing: Packaged IC tested at full speed, temperature extremes, and burn-in for reliability screening.

Photolithography Process Steps (one layer cycle):

```
Step 1: OXIDATION
  Wafer heated to 900-1200°C in O2/H2O atmosphere
  → SiO2 grows on silicon surface (~100nm thick)
   ┌──────────────────────────────┐
   │   SiO2 (oxide layer)         │  ← grown oxide
   │   Silicon Wafer              │
   └──────────────────────────────┘


Step 2: PHOTORESIST APPLICATION (Spin Coating)
  Liquid photoresist spun onto wafer at high RPM → uniform thin film
   ┌──────────────────────────────┐
   │▒▒▒ Photoresist (PR) ▒▒▒▒▒▒▒▒▒│  ← ~1µm thick
   │   SiO2                       │
   │   Silicon Wafer              │
   └──────────────────────────────┘


Step 3: SOFT BAKE  →  ~90°C, evaporates solvent, hardens PR

Step 4: MASK ALIGNMENT + UV EXPOSURE
  Photomask (quartz + chrome pattern) aligned over wafer
  UV light exposes through clear areas of mask
   ┌──────────────────────────────┐
   │▐▐ Chrome ▐▐│  Clear  │▐▐▐▐▐▐│  ← Photomask
   │──────────────────────────────│  UV light ↓
   │▒▒▒▒▒▒▒▒▒▒▒▒▒▒ │ exposed │▒▒▒▒▒▒│  ← PR (exposed = soluble)
   │   SiO2                       │
   └──────────────────────────────┘


Step 5: DEVELOPMENT
  Developer dissolves exposed positive resist → pattern revealed
   ┌──────────────────────────────┐
   │▒▒▒▒▒▒▒▒▒▒▒▒▒▒ │         │▒▒▒▒▒▒│  ← remaining PR (hardened)
   │   SiO2      │  open   │      │  ← oxide exposed
   └──────────────────────────────┘


Step 6: HARD BAKE  →  ~120°C, further hardens remaining resist
```

```
Step 7: ETCHING
  HF acid etches exposed SiO2; resist protects other areas
  ┌──────────────────────────────┐
  │▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ |          |▒▒▒▒▒▒▒|  ← resist still present
  |    SiO2      |  window |       |  ← oxide removed
  |    Silicon   |▉▉▉▉▉▉▉▉▉|       |  ← silicon exposed
  └──────────────────────────────┘


Step 8: RESIST STRIPPING
  Remaining photoresist removed with solvent/plasma
  ┌──────────────────────────────┐
  |    SiO2      |  window |       |  ← final patterned oxide
  |    Silicon Wafer               |
  └──────────────────────────────┘


  → Repeat all 8 steps for each subsequent layer (doping, gate, metal...)
```

Q2. Discuss the advantages and disadvantages of full-custom IC technology. What types of products use full-custom ICs?	**2072 Ashwin (Regular) | 2078 Chaitra (Back) | 2074 Jestha (Back) [3+5]**

ANSWER

Full-Custom IC Technology:

In full-custom (VLSI) design, every transistor, every wire, and every cell is designed manually at the mask layout level by experienced VLSI designers. Nothing is pre-placed or pre-characterized — every dimension is optimized for the specific application.

Advantages	Disadvantages
Maximum performance: Transistors sized and placed for minimum delay; critical paths fully optimized	Very high NRE (Non-Recurring Engineering) cost: Full mask set for advanced node (7nm) costs $20M–$100M+
Minimum chip area: No wasted silicon — every transistor is optimally placed and sized	Extremely long design time: Months to years for complex chips; large expert team required
Minimum power consumption: Transistor sizing minimizes switching power	Requires highly skilled VLSI designers: Rare and expensive expertise needed
Best for analog and mixed-signal circuits: Precise matching, custom layout critical for analog performance	High risk: A single mask error requires expensive re-fabrication (re-spin)
Competitive advantage: Unique performance impossible with semi-custom	Only economical at very high volume: NRE amortized only over millions of units
Full control over every design decision: No library constraints	Longer time-to-market compared to FPGA/standard cell

Products Using Full-Custom ICs:

- Microprocessors and GPUs: Intel Core, AMD Ryzen, NVIDIA A100 — critical datapaths hand-optimized.
- Mobile SoCs: Apple A-series, Qualcomm Snapdragon — billions of units justify NRE.
- Memory ICs: DRAM and NAND Flash cells are full-custom for maximum density (Samsung, Micron).
- Analog ICs: Precision ADCs, PLLs, RF circuits — layout matching and parasitics critical.
- High-speed SerDes: 100 Gbps+ transceivers — only custom design meets timing.

Q3. What are the merits and demerits of full-custom and semi-custom IC technology? Explain gate-array and standard-cell semi-custom IC technologies.	**2081 Chaitra (Regular) | 2079 Chaitra (Back) | 2073 Bhadra (Back) [4+4]**

ANSWER

Feature	Full-Custom	Semi-Custom (ASIC)
Performance	Highest — every transistor optimized	Good — slightly less than full-custom
Design Time	Very long (months–years)	Moderate (weeks–months)
NRE Cost	Very high (full mask set, millions)	Moderate (partial or full masks, $500K–$5M)
Flexibility	Maximum optimization freedom	Limited to library cells or pre-placed transistors
Chip Area	Minimum	10–30% larger than full-custom
Designer skill	Expert VLSI layout designers	RTL/synthesis engineers (more common)
Volume needed	Very high (millions+)	Medium to high (thousands to millions)
Examples	Intel CPU, Apple A-chip	Telecom ASICs, networking chips

Gate-Array Semi-Custom IC Technology:

A gate-array IC is prefabricated with a regular array of uncommitted transistor cells on a silicon wafer up to — but not including — the metal interconnect layers. Wafers are stockpiled and only the metal mask layers are customized per design.

```
Gate-Array Structure:
┌──────────────────────────────────────────────────┐
│  I/O Pads                                        │
│  ┌──────────────────────────────────────────┐    │
│  │ Row: [Cell][Cell][Cell]...[Cell]          │    │
│  │      ← Routing Channel →                 │    │
│  │ Row: [Cell][Cell][Cell]...[Cell]          │    │
│  │      ← Routing Channel →                 │    │
```

```
│   │ Row: [Cell][Cell][Cell]...[Cell]          │   │
│   └──────────────────────────────────────────┘   │
└──────────────────────────────────────────────────┘

  All cells pre-fabricated (transistors placed)
  Only METAL layers are custom — fast turnaround
```

- Advantage: Short time-to-market (days to weeks once design is complete); lower cost than full-custom since most fabrication steps are shared (pre-fabricated wafer stockpile).
- Disadvantage: Transistors are pre-placed in fixed positions — cannot optimize cell sizes. Some cells may not be used (wasted area). Routing channels reduce density.
- Types: Channeled gate array (routing between rows), channel-less/Sea-of-Gates (denser, routing over cells).

Standard Cell Semi-Custom IC Technology:

Standard cells are pre-designed, pre-characterized logic cells (NAND2, NOR2, DFF, adder, MUX) with fixed height but variable width, stored in a technology library.

```
Standard Cell Design Flow:
  HDL (VHDL/Verilog) → [Synthesis] → Netlist (gates)
       → [Floorplanning] → [Placement] → [Clock Tree Synthesis]
       → [Routing] → [STA] → [DRC/LVS] → GDSII → Fabrication

Standard Cell Row Layout:
┌──────────────────────────────────────────────┐
│ NAND │ INV │ DFF  │ NOR │ MUX │ BUF │ AND  │  ← row of cells
├──────────────────────────────────────────────┤  VDD/GND rails
│ DFF  │ XOR │ NAND │ INV │ DFF │ OR  │ NAND │  ← next row
└──────────────────────────────────────────────┘
  Variable width, fixed height — rows abutted, VDD/GND shared
```

- Advantage: Better area and performance than gate arrays (cells individually optimized). EDA tools (Synopsys, Cadence) automate synthesis, placement, and routing. Most widely used ASIC methodology today.
- Disadvantage: Full mask set required — higher NRE than gate arrays. Limited to available cell library functions.
- Used by: Qualcomm, MediaTek, ARM, Broadcom — most commercial ASICs.

Q4. Describe PLD (Programmable Logic Device) IC technology. Explain SPLD, CPLD, and FPGA with diagrams.	**2078 Poush (Back) | 2073 Bhadra (Back) | 2076 Chaitra (Back) [8]**

ANSWER

PLD (Programmable Logic Device) Overview:

PLDs are ICs that can be configured by the designer after manufacture to implement any desired digital logic function — no custom fabrication required. They are programmed using a hardware description language (VHDL/Verilog) and a software toolchain.

1. SPLD — Simple PLD (PAL/GAL):

SPLDs contain a programmable AND array and a fixed OR array, or both programmable AND and OR arrays. They implement sum-of-products (SOP) Boolean expressions.

```
SPLD (PAL) Internal Structure:

Inputs ──►┌──────────────────────────────┐
 A ──────►│  Programmable AND Array      │──► Product Terms
 B ──────►│  (fuse/antifuse matrix)      │──► (AND gates)
 C ──────►│                              │──► P1 = A·B'
          └──────────────────────────────┘    P2 = A'·C
                        │                     P3 = B·C
                        ▼
          ┌──────────────────────────────┐
          │  Fixed OR Array              │──► F1 = P1+P2
          │                              │──► F2 = P2+P3
          └──────────────────────────────┘
                        │
          ┌──────────────────────────────┐
          │  Output Logic (optional FF)  │──► Outputs
          └──────────────────────────────┘

Programming: Blow fuses in AND array to select inputs for each product
term
Capacity: Tens to hundreds of gates
Examples: PAL16L8, GAL22V10
```

2. CPLD — Complex PLD:

A CPLD consists of multiple SPLD-like macrocell blocks connected by a programmable interconnect matrix (PIM). Offers much higher capacity than SPLDs while maintaining predictable timing.

```
CPLD Structure:
┌──────────────────────────────────────────────────────┐
│                        CPLD                          │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐           │
│  │ Macro-   │  │ Macro-   │  │ Macro-   │           │
│  │ cell     │  │ cell     │  │ cell     │           │
│  │ Block 1  │  │ Block 2  │  │ Block 3  │           │
│  │(AND-OR   │  │(AND-OR   │  │(AND-OR   │           │
│  │ array)   │  │ array)   │  │ array)   │           │
│  └──────────┘  └──────────┘  └──────────┘           │
│       └──────────────┴──────────────┘                │
│            Programmable Interconnect Matrix (PIM)    │
│                  (switch matrix)                     │
│  I/O Blocks ◄──────────────────────────────────────  │
└──────────────────────────────────────────────────────┘
```

```
Capacity: Hundreds to thousands of gates
Programming: EEPROM or Flash cells (non-volatile, retains config at power-
off)
Timing: Predictable (pin-to-pin delay specified, unlike FPGA)
Examples: Altera MAX series, Xilinx XC9500, Lattice ispMACH
```

3. FPGA — Field Programmable Gate Array:

FPGAs are the most powerful and flexible PLDs. They consist of a large array of configurable logic blocks (CLBs), connected by a programmable routing network, with I/O blocks around the perimeter. Modern FPGAs also include hard IP blocks (multipliers, RAM, DSP slices, PCIe, Ethernet MACs, ARM cores).

```
FPGA Structure:
┌──────────────────────────────────────────────────────────┐
│  I/O   │                    FPGA                  │  I/O   │
│ Blocks │ ┌─────┐   ┌─────┐   ┌─────┐   ┌─────┐   │ Blocks │
│        │ │ CLB │──│ CLB │──│ CLB │──│ CLB │   │        │
│        │ └─────┘   └─────┘   └─────┘   └─────┘   │        │
│        │     │     │     │     │     │     │     │        │
│        │     │  Prog. │  Route │  Matrix │     │        │
│        │ ────┴────  ────┴────  ────┴────  ────┴────  │        │
│        │ ┌─────┐   ┌─────┐   ┌─────┐   ┌─────┐   │        │
│        │ │ CLB │   │ CLB │   │ DSP │   │BRAM │   │        │
│        │ └─────┘   └─────┘   └─────┘   └─────┘   │        │
│        │    Programmable Routing Network       │        │
└──────────────────────────────────────────────────────────┘


CLB (Configurable Logic Block) contains:
  • LUT (Look-Up Table): 4-6 input truth table → implements any Boolean fn
  • Flip-Flop (D-FF): Sequential logic / registers
  • MUX: Connects LUT output to FF or directly to routing

Configuration: SRAM-based (most common — reprogrammable, volatile — needs
  bitstream reload at power-on) or Flash/Antifuse (non-volatile)
Capacity: Thousands to millions of LUTs
Examples: Xilinx Artix-7, Virtex UltraScale+; Intel Cyclone, Stratix
```

SPLD vs CPLD vs FPGA Comparison:

Property	SPLD (PAL/GAL)	CPLD	FPGA
Capacity	Small (tens of gates)	Medium (hundreds–thousands)	Large (thousands–millions of LUTs)
Architecture	AND-OR array	Multiple AND-OR macrocell blocks + PIM	LUT array + programmable routing + hard IP
Configuration	Fuse/EEPROM	EEPROM/Flash (non-volatile)	SRAM (volatile) or Flash/Antifuse
Timing	Predictable	Predictable pin-to-pin delay	Less predictable (routing dependent)

Power	Low	Low-Medium	Medium-High (many SRAM bits)
Non-volatile	Yes (most)	Yes	No (SRAM) / Yes (Flash/Antifuse)
Reprogrammable	Yes (GAL)	Yes	Yes (SRAM, many times)
Examples	PAL16L8, GAL22V10	MAX7000, XC9500	Artix-7, Cyclone V, Virtex-7
Best used for	Simple glue logic	Interface, control logic	Complex DSP, prototyping, large designs

Q5. Describe the design steps involved for Full-Custom, Semi-Custom, and PLD IC technologies.	**2078 Poush (Back) | 2074 Bhadra (Back) | 2076 Baisakh (Back) [8]**

ANSWER

Full-Custom IC Design Steps:

```
1. System Specification → 2. Architecture Design → 3. Circuit Design
     → 4. Manual Layout → 5. DRC → 6. LVS → 7. Parasitic Extraction
     → 8. Post-Layout Simulation → 9. GDSII Generation → 10. Fabrication
```

1. System Specification: Define functional requirements, performance targets (speed, power), area budget, and process technology node.
2. Architecture Design: Partition into functional blocks (datapath, control, memory, I/O). Define interface signals.
3. Circuit Design: Design transistor-level schematics. Size each MOSFET (W/L ratio) for required drive strength, speed, and power. Simulate using SPICE.
4. Manual Layout: Draw every mask layer (active, poly, n-well, contacts, metal1, metal2...) using Virtuoso/Laker. Every transistor is manually placed and sized.
5. Design Rule Check (DRC): Verify layout satisfies fabrication constraints (minimum width, spacing, overlap rules) for the chosen process node.
6. Layout vs. Schematic (LVS): Extract netlist from layout and compare with original schematic — must match exactly.
7. Parasitic Extraction and Post-Layout Simulation: Extract resistances and capacitances from layout; re-simulate timing to verify performance with actual parasitics.
8. GDSII Generation: Generate GDSII (Graphic Database System II) file — the standard format for photomask fabrication.
9. Fabrication, Packaging, and Testing: (See Q1.)

Semi-Custom (Standard Cell ASIC) Design Steps:

```
RTL (VHDL/Verilog) → [Synthesis] → Gate Netlist
  → [Floorplan] → [Placement] → [CTS] → [Routing]
  → [STA] → [DRC/LVS] → [GDSII] → Fabrication
```

1. RTL Design (HDL Coding): Write Register-Transfer Level description in VHDL or Verilog. Simulate with testbench for functional verification.
2. Logic Synthesis: Synthesis tool (Synopsys Design Compiler) maps RTL to standard cells from the foundry's cell library. Optimizes for timing, area, or power.
3. Gate-Level Simulation: Verify synthesized netlist against RTL simulation results. Check for synthesis issues.
4. Floorplanning: Define chip boundary; plan placement of large blocks (memories, analog blocks); place I/O pads; define power grid (VDD/GND rings and stripes).
5. Placement: Place all standard cells in rows within floorplan. Tools minimize wire lengths to meet timing constraints.
6. Clock Tree Synthesis (CTS): Build a balanced clock distribution network to minimize clock skew between all flip-flops.
7. Routing: Connect all cell pins with metal wires following design rules. Fill remaining area with filler cells.
8. Static Timing Analysis (STA): Verify all setup and hold timing constraints are met with actual wire delays (using Synopsys PrimeTime).
9. DRC/LVS, GDSII generation, Fabrication, and Testing.

PLD (FPGA) Design Steps:

```
HDL/Schematic → [Synthesis] → [Implementation: Place&Route]
  → [Timing Analysis] → [Bitstream Generation] → [JTAG Programming] → Test
```

1. Design Entry: Write HDL (VHDL/Verilog) or use schematic capture / IP block integration.
2. Functional Simulation: Verify RTL behavior with testbench before synthesis.
3. Synthesis: Xilinx Vivado / Intel Quartus synthesizes HDL into an FPGA netlist (LUTs, FFs, BRAMs, DSPs).
4. Implementation (Place and Route): Tool places LUTs/FFs onto specific FPGA sites and routes connections through the programmable routing fabric.
5. Timing Analysis: Verify all timing constraints (setup/hold, clock period) are met with actual FPGA routing delays.
6. Bitstream Generation: Create device-specific binary configuration file (.bit for Xilinx, .sof for Intel).
7. Device Programming: Download bitstream to FPGA via JTAG cable (Xilinx Platform Cable, Intel USB-Blaster). Configuration stored in SRAM (needs reload at power-on) or external SPI Flash (auto-loads).
8. Hardware Testing: Verify design on target board using logic analyzer, oscilloscope, or embedded logic analyzer (ILA/SignalTap).

Q6. Define photolithography and explain its types in the context of IC manufacturing.	**2080 Ashwin (Back) | 2079 Ashwin (Back) | 2077 Chaitra (Back) [4+4]**

ANSWER

Photolithography:

Photolithography (also called optical lithography or microlithography) is the process of transferring geometric patterns from a photomask to the surface of a semiconductor wafer using light exposure and light-sensitive photoresist chemistry. It is the most critical and repeated step in IC fabrication — performed 20–30 times per chip to define each layer.

Types of Photolithography:

1. Contact Lithography: The photomask is placed in direct physical contact with the photoresist-coated wafer during exposure.

- Resolution: ~1–2 μm (limited by diffraction and mask wear).
- Advantage: Simple, inexpensive equipment.
- Disadvantage: Physical contact damages mask quickly; defects from dust particles; poor yield. Used only in research and low-volume old-process fabrication.

2. Proximity Lithography: A small gap (10–50 μm) is maintained between mask and wafer during exposure.

- Resolution: ~2–5 μm (worse than contact due to diffraction across gap).
- Advantage: Eliminates mask wear problem compared to contact lithography.
- Disadvantage: Lower resolution due to diffraction across the gap. Not suitable for modern IC features.

3. Projection Lithography (Most Important): A reduction optical system projects the mask image (typically 4× or 5× reduced) onto the wafer. Used in all modern IC manufacturing.

- Resolution: Sub-100nm with advanced techniques (DUV, immersion, multiple patterning).
- Sub-types: Stepper (exposes one die at a time, then steps to next position) and Scanner (slit scans across die — better uniformity, used in modern fabs).
- Light sources: i-line (365 nm), KrF (248 nm), ArF (193 nm DUV — most common today), ArF immersion (193 nm with water immersion — achieves ~40 nm features).
- Advantage: High resolution, mask reused thousands of times, mass production.

4. Electron Beam (E-Beam) Lithography: A focused electron beam directly writes patterns onto the resist without a mask.

- Resolution: Sub-10 nm (highest resolution available, ~1–5 nm in research).
- Advantage: No mask needed — direct write; used for mask fabrication itself and research.
- Disadvantage: Extremely slow (serial write, not parallel) — cannot be used for high-volume wafer production.

5. EUV (Extreme Ultraviolet) Lithography: Uses 13.5 nm wavelength light generated by a high-energy laser hitting a tin droplet target. All optics are reflective mirrors (no lenses work at EUV wavelengths).

- Resolution: Sub-5 nm features (currently used for 3nm, 2nm nodes by TSMC and Samsung).

- Advantage: Enables continued Moore's Law scaling; eliminates multiple patterning complexity of DUV.
- Disadvantage: Extremely expensive equipment ($150M+ per scanner); requires near-perfect vacuum; low light source power limits throughput.

Type	Wavelength	Resolution	Volume Production	Use Case
Contact	UV (365nm)	~1–2 µm	No (mask wear)	Research, old processes
Proximity	UV (365nm)	~2–5 µm	No	Research only
Projection (DUV)	193nm (ArF)	40–130nm	Yes — mainstream	14nm–130nm nodes
Immersion DUV	193nm + H2O	~40nm (single)	Yes — mainstream	7nm–40nm (with multi-patterning)
E-Beam	Electrons (~1nm)	<10nm	No (too slow)	Mask writing, research
EUV	13.5nm	<5nm	Yes — cutting edge	3nm, 2nm, 1nm nodes

Q7. Compare Full-Custom, Semi-Custom (ASIC), and PLD (FPGA) IC technologies from an embedded system designer's perspective.	**2081 Chaitra (Regular) | 2080 Chaitra (Back) | 2074 Chaitra (Back) [8]**

ANSWER

Property	Full-Custom (VLSI)	Semi-Custom (ASIC)	PLD (FPGA/CPLD)
Performance	Highest — manually optimized	High — synthesis-optimized cells	Moderate — routing overhead
Power	Lowest — optimized transistors	Low-moderate	Higher (SRAM config overhead)
Chip Area	Smallest	Moderate (10–30% more than full-custom)	Largest (LUT overhead)
NRE Cost	Very high ($5M–$100M+)	Moderate ($500K–$5M)	Zero (off-shelf chip)
Unit Cost	Lowest (at high volume)	Low (at medium-high volume)	Higher per unit
Design Time	Months to years	Weeks to months	Days to weeks
Time-to-Market	Longest	Moderate	Fastest
Flexibility	None after tapeout	None after tapeout	Fully reprogrammable in field
Volume Economics	Best at millions+ units	Best at thousands–millions units	Best at small–medium volumes
Tools	Virtuoso, Calibre (expensive)	Synopsys, Cadence (expensive)	Vivado, Quartus (free/low cost)

Non-volatile	N/A	N/A	SRAM=No; Flash/Antifuse=Yes
Examples	CPU, GPU, DRAM	Networking chips, baseband modems	Prototyping, industrial ctrl, video processing

Selection Guidelines for Embedded System Designers:

Choose Full-Custom when: Product volume is extremely high (10M+ units/year), performance is critical, power budget is very tight, and development budget is large (smartphones, high-end processors, memory).

Choose Semi-Custom (ASIC) when: Volume justifies NRE (100K+ units), design is finalized and won't change, better performance/power than FPGA needed (telecom, automotive, IoT at scale).

Choose FPGA when: Design needs flexibility (field updates, algorithm changes), time-to-market is critical, volume is low-to-medium, or for prototyping before ASIC tapeout. Also when design is too large for CPLD but ASIC NRE not justified.

Choose CPLD when: Simple glue logic, interface bridging, power-on configuration logic — small designs needing non-volatile, fast, predictable timing.

Q8. Explain CMOS technology. Design CMOS NAND gate and CMOS NOR gate with circuit diagrams and truth tables.	**2079 Ashwin (Back) | 2078 Baishakh (Back) | 2076 Baisakh (Back) [3+5]**

ANSWER

CMOS (Complementary Metal-Oxide-Semiconductor) Technology:

CMOS is the dominant IC fabrication technology. It uses complementary pairs of NMOS (n-channel) and PMOS (p-channel) MOSFETs. The key property is that at steady state, one transistor is always OFF — no direct DC path from VDD to GND — giving near-zero static power consumption.

- NMOS transistor: ON when gate = HIGH (VGS > Vth). Connects output to GND (pull-down network, PDN).
- PMOS transistor: ON when gate = LOW (VGS < −|Vth|). Connects output to VDD (pull-up network, PUN).
- Static power: Near zero (only leakage current when both transistors off simultaneously).
- Dynamic power: $P = \alpha \cdot C \cdot VDD^2 \cdot f$ — proportional to switching frequency, load capacitance, and VDD^2.

CMOS NAND Gate (2-input):

```
                VDD
                 |
            ┌────┴────┐
A ──────────┤ PMOS P1 │  (A=0 → P1 ON)
            └────┬────┘
```

```
                 │
            ┌────┴────┐
B ──────────┤ PMOS P2 │  (B=0 → P2 ON)
            └────┬────┘
                 │
                 ├──────────────── Y = (A·B)' output
                 │
            ┌────┴────┐
A ──────────┤ NMOS N1 │  (A=1 → N1 ON)
            └────┬────┘
                 │
            ┌────┴────┐
B ──────────┤ NMOS N2 │  (B=1 → N2 ON)
            └────┬────┘
                 │
               GND

PUN: P1 and P2 in PARALLEL — output HIGH if A=0 OR B=0
PDN: N1 and N2 in SERIES  — output LOW  if A=1 AND B=1
```

A	B	N1 (NMOS)	N2 (NMOS)	P1 (PMOS)	P2 (PMOS)	Y = (AB)'
0	0	OFF	OFF	ON	ON	1 (VDD through P1 and P2)
0	1	OFF	ON	ON	OFF	1 (VDD through P1)
1	0	ON	OFF	OFF	ON	1 (VDD through P2)
1	1	ON	ON	OFF	OFF	0 (GND through N1 and N2)

CMOS NOR Gate (2-input):

```
               VDD
                 │
            ┌────┴────┐
A ──────────┤ PMOS P1 │  (A=0 → P1 ON)
            └────┬────┘
                 │
            ┌────┴────┐
B ──────────┤ PMOS P2 │  (B=0 → P2 ON)
            └────┬────┘
                 │
                 ├──────────────── Y = (A+B)' output
                 │              ┌────┴────┐
            ┌────┴────┐         │ NMOS N2 ├──── B
A ──────────┤ NMOS N1 │         └────┬────┘
            └────┬────┘              │
                 └──────────────────┘
                       GND
```

```
PUN: P1 and P2 in SERIES   — output HIGH only if A=0 AND B=0
PDN: N1 and N2 in PARALLEL — output LOW  if A=1 OR  B=1
```

A	B	N1 (NMOS)	N2 (NMOS)	P1 (PMOS)	P2 (PMOS)	Y = (A+B)'
0	0	OFF	OFF	ON	ON	1 (VDD through P1 and P2 in series)
0	1	OFF	ON	ON	OFF	0 (GND through N2)
1	0	ON	OFF	OFF	ON	0 (GND through N1)
1	1	ON	ON	OFF	OFF	0 (GND through N1 or N2)

Key Design Rule for CMOS Gates:

- PUN (Pull-Up Network) is the dual of PDN (Pull-Down Network): Series in PDN → Parallel in PUN; Parallel in PDN → Series in PUN.
- NAND gate: PDN = series NMOS (both must be ON to pull down) → PUN = parallel PMOS (either can pull up).
- NOR gate: PDN = parallel NMOS (either pulls down) → PUN = series PMOS (both must be ON to pull up).
- PMOS transistors must be wider (typically 2–3× W) than NMOS for equal drive strength due to lower hole mobility.

Q9. Draw and explain the CMOS inverter circuit. What is propagation delay and how does it affect IC performance?

Model Question | 2075 Chaitra (Back) [4+4]

ANSWER

CMOS Inverter:

```
            VDD
             |
        ┌────┴────┐
A ──────┤  PMOS   │   (when A=0: PMOS ON, NMOS OFF → Y=VDD=1)
        └────┬────┘
             |
             ├──────────── Y = A'   (output)
             |
        ┌────┴────┐
A ──────┤  NMOS   │   (when A=1: NMOS ON, PMOS OFF → Y=GND=0)
        └────┬────┘
             |
            GND

Truth table: A=0 → Y=1;  A=1 → Y=0   (NOT gate)
Static power: ~0 (one transistor always OFF — no VDD-to-GND path)
Dynamic power: P = C·VDD²·f  (charges/discharges load capacitance C)
```

Operation:

- Input A = LOW (0V): PMOS gate-source voltage VGS = −VDD < −|Vthp| → PMOS ON. NMOS VGS = 0 < Vthn → NMOS OFF. Output Y charged to VDD = logic HIGH.
- Input A = HIGH (VDD): NMOS VGS = VDD > Vthn → NMOS ON. PMOS VGS = 0 > −|Vthp| → PMOS OFF. Output Y discharged to GND = logic LOW.
- At any stable state: One transistor ON, one OFF — no static DC current path. Near-zero static power.

Propagation Delay:

Propagation delay (tp) is the time from when the input transitions 50% to when the output transitions 50%. It is the fundamental measure of logic gate speed.

Parameter	Symbol	Definition
Rise time	tr	Time for output to transition from 10% to 90% of VDD
Fall time	tf	Time for output to fall from 90% to 10% of VDD
High-to-Low delay	tpHL	Input rises 50% → output falls to 50%
Low-to-High delay	tpLH	Input falls 50% → output rises to 50%
Propagation delay	tp	Average: tp = (tpHL + tpLH) / 2

Factors affecting propagation delay:

- Load capacitance (CL): tp □ CL — larger fan-out or wire capacitance → slower. tp = CL × VDD / (2 × Ion).
- Supply voltage (VDD): Lower VDD → less drive current → slower switching. tp ∝ 1/VDD.
- Transistor size (W/L): Wider transistors → more current → faster. But larger area → more capacitance.
- Temperature: Higher temperature → lower carrier mobility → slower switching.
- Process node: Smaller transistors (7nm vs 28nm) → shorter channels → faster switching, lower capacitance.
- Impact on IC performance: Maximum clock frequency fmax ≈ 1/(tp × N_stages) where N_stages is pipeline depth. Minimizing propagation delay is the primary goal of VLSI optimization.

Summary: Topics Covered in Chapter 8

S.N.	Topic	Exam(s) Asked	Marks
1	IC Manufacturing Steps (13 steps, full process)	2082 Kartik, 2081 Ashwin, 2080 Ashwin, 2075 Bhadra (annual)	3–8
2	Photolithography Process Steps (8 steps with ASCII diagrams)	Almost every exam — most important topic	5–8
3	Types of Photolithography (Contact/Proximity/Projection/E-Beam/EUV)	2080 Ashwin, 2079 Ashwin, 2077 Chaitra	4+4
4	Full-Custom IC — Advantages and Disadvantages	2072 Ashwin, 2078 Chaitra, 2074 Jestha	3+5
5	Semi-Custom IC — Full-Custom vs Semi-Custom comparison table	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4+4
6	Gate-Array Semi-Custom — structure + advantages/disadvantages	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4
7	Standard Cell Semi-Custom — design flow + advantages	2081 Chaitra, 2079 Chaitra, 2073 Bhadra	4
8	PLD — SPLD/PAL structure with diagram	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
9	CPLD — structure, macrocell blocks, PIM diagram	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
10	FPGA — CLB/LUT structure, SRAM configuration, hard IP	2078 Poush, 2073 Bhadra, 2076 Chaitra	4
11	SPLD vs CPLD vs FPGA full comparison table	2078 Poush, 2073 Bhadra	4+4
12	Design Steps: Full-Custom (9 steps)	2078 Poush, 2074 Bhadra, 2076 Baisakh	8
13	Design Steps: Semi-Custom ASIC (10 steps)	2078 Poush, 2074 Bhadra	8
14	Design Steps: PLD/FPGA (8 steps)	2078 Poush, 2073 Bhadra	8
15	Full-Custom vs Semi-Custom vs PLD — full comparison table	2081 Chaitra, 2080 Chaitra, 2074 Chaitra	8
16	CMOS Technology + NAND gate (circuit + truth table)	2079 Ashwin, 2078 Baishakh, 2076 Baisakh (annual)	3+5
17	CMOS NOR gate (circuit + truth table)	2079 Ashwin, 2078 Baishakh, 2076 Baisakh	3+5
18	CMOS Inverter + Propagation Delay	2075 Chaitra, Model Q	4+4